

The ratio of three quantities

Chaining two ratios into one, and sharing among three.

LESSON 47–49

3 × 40 MIN

MATEMATIKA 6, §11

S7 12

LESSON CLOCK

25'

30'

30'

25'

10

Sharing among three	25 min
Chaining two ratios	30 min
From one quantity to all three	30 min
Mixtures	25 min
Homework	10 min

LEARNING OBJECTIVES

- Share a quantity among three in a given ratio.
- Combine $a : b$ and $b : c$ into $a : b : c$.
- Find all three quantities from one of them.
- Solve mixture problems with three ingredients.

TEXTBOOK

National	pp. 131–140
Cambridge	Stage 7, pp. 118–125
Workbook	Exercise 12.4

01

Explanation

Sharing among three

Nothing new: add all three parts, find one part, multiply out.

AMOUNT	RATIO	PARTS	SHARES
300	1 : 2 : 3	6	50, 100, 150
700	1 : 2 : 4	7	100, 200, 400
540	2 : 3 : 4	9	120, 180, 240
480	3 : 5 : 8	16	90, 150, 240

THE CHECK IS THE SAME

All three shares must add to the original amount. With three numbers to add there is more to go wrong, so the check matters more, not less.

Chaining two ratios

Given $A : B = 2 : 3$ and $B : C = 4 : 5$, the two B s must be made equal before the ratios can be joined.

STEP		WORKING
the two ratios		$A : B = 2 : 3, B : C = 4 : 5$
make the B s match — LCM of 3 and 4 is 12		$A : B = 8 : 12, B : C = 12 : 15$
join them		$A : B : C = 8 : 12 : 15$
$A : B$	$B : C$	$A : B : C$
$2 : 3$	$3 : 4$	$2 : 3 : 4$
$2 : 3$	$4 : 5$	$8 : 12 : 15$
$1 : 2$	$3 : 7$	$3 : 6 : 14$

ONLY THE SHARED QUANTITY IS MATCHED

Scaling $A : B$ by 4 and $B : C$ by 3 changes the numbers but not the ratios — that is exactly why it is allowed. Scaling only one of the two would break them.

From one quantity to all three

RATIO	GIVEN	ONE PART	ALL THREE
$2 : 3 : 5$	the first is 12	6	12, 18, 30
$1 : 4 : 6$	the second is 20	5	5, 20, 30
$3 : 4 : 5$	the third is 35	7	21, 28, 35
$2 : 3 : 4$	the total is 180	20	40, 60, 80

ALWAYS ONE PART FIRST

Whatever is given — a share, the total, or a difference — the first line of working is the value of one part. Every three-quantity problem then finishes in one more line.

Mixtures

PROBLEM	WORKING	ANSWER
concrete $1 : 2 : 4$ using 150 kg of cement	one part = 150	1050 kg in all
a salad $3 : 2 : 1$ of 600 g	one part = 100	$300, 200, 100$ g
brass $7 : 3$ copper to zinc in 500 g	one part = 50	350 g copper
a mix $2 : 3 : 5$ with 60 g of the largest part	one part = 12	120 g in all

READ WHICH PART THE GIVEN AMOUNT IS

“Using 150 kg of cement” names the 1 ; “with 60 g of the largest part” names the 5 . The same number leads to very different totals depending on which part it is.

02

Worked examples

EXAMPLE 1 Share 540 in the ratio $2 : 3 : 4$.

1

$2 + 3 + 4 = 9$ parts.

2

$540 \div 9 = 60$ for one part.

3

$120, 180$ and 240 .
Check: they total 540 ✓

ANSWER

$120, 180, 240$

EXAMPLE 2 If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.

① The Bs are 3 and 4; their LCM is 12.

② Scale the first by 4: $8 : 12$.

③ Scale the second by 3: $12 : 15$.

④ $A : B : C = 8 : 12 : 15$

ANSWER

$8 : 12 : 15$

EXAMPLE 3 Concrete is $1 : 2 : 4$ cement, sand and gravel. A builder uses 150 kg of cement. What is the total mass?

① Cement is 1 part, so one part is 150 kg.

② Total parts: 7.

③ $7 \cdot 150 = 1050$ kg.

ANSWER

1050 kg

03 Interactive model

Give the class two ratios sharing a middle quantity and ask them to join them without being shown how; most reach the LCM idea themselves.

Matching the shared quantity

DENOMINATOR	FACTORISED	EXTRA FACTOR NEEDED
3	<i>B in the first ratio</i>	$\times 4$
4	<i>B in the second</i>	$\times 3$

LOWEST COMMON DENOMINATOR

$$A : B : C = 8 : 12 : 15$$

The LCM of 3 and 4 is 12.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Three-part ratio	Uch hadli nisbat	Отношение трёх величин
To combine ratios	Nisbatlarni birlashtirish	Объединить отношения
Common term	Umumiy had	Общий член
Equivalent ratio	Teng nisbat	Равное отношение
Share	Ulush	Доля
Mixture	Aralashma	Смесь
Total parts	Jami ulushlar	Всего частей
Scaling	Miqyoslash	Масштабирование

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

300 in $1 : 2 : 3$ gives:

50, 100, 150

60, 100, 140

100, 100, 100

30, 60, 210

To join $A : B$ and $B : C$ you make equal:

the As

the Bs

the Cs

the totals

$2 : 3$ and $4 : 5$ join to:

$2 : 3 : 5$

$8 : 12 : 15$

$2 : 12 : 5$

$6 : 12 : 20$

Ratio $2 : 3 : 5$ with the first 12 gives the third:

18

24

30

36

Concrete $1 : 2 : 4$ with 150 kg cement totals:

450 kg

750 kg

1050 kg

1200 kg

The first line of working is always:

the total

one part

the ratio

the check

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Share 300 in 1 : 2 : 3

ANSWER 50, 100, 150

2. Share 700 in 1 : 2 : 4

ANSWER 100, 200, 400

3. Share 540 in 2 : 3 : 4

ANSWER 120, 180, 240

4. Share 480 in 3 : 5 : 8

ANSWER 90, 150, 240

5. The parts of 2 : 3 : 5

ANSWER 10

6. Ratio 2 : 3 : 5, first 12: one part

ANSWER 6

7. And all three

ANSWER 12, 18, 30

Medium · the standard question

7 PROBLEMS

1. $A : B = 2 : 3, B : C = 3 : 4$

ANSWER $2 : 3 : 4$

2. $A : B = 2 : 3, B : C = 4 : 5$

ANSWER $8 : 12 : 15$

3. $A : B = 1 : 2, B : C = 3 : 7$

ANSWER $3 : 6 : 14$

4. Ratio $1 : 4 : 6$, second 20

ANSWER $5, 20, 30$

5. Ratio $3 : 4 : 5$, third 35

ANSWER $21, 28, 35$

6. Concrete $1 : 2 : 4$ with 150 kg cement

ANSWER 1050 kg

7. A salad $3 : 2 : 1$ of 600 g

ANSWER $300, 200, 100$ g

Hard · stretch and reason

7 PROBLEMS

1. Brass 7 : 3 in 500 g: the copper

ANSWER 350 g

2. A mix 2 : 3 : 5 with 60 g of the largest part: the total

ANSWER 120 g

3. $A : B = 3 : 4$, $B : C = 6 : 7$: find $A : B : C$

ANSWER 9 : 12 : 14

4. Share 2 400 in 1 : 3 : 4

ANSWER 300, 900, 1200

5. Ratio 2 : 3 : 4, the difference between largest and smallest is 30

ANSWER 30, 45, 60

6. If $A : B = 2 : 5$ and $B : C = 10 : 3$, find $A : C$

ANSWER 4 : 3

7. Why must the shared quantity be matched before joining?

ANSWER Otherwise the two ratios measure B in different units

07 Homework

Homework — 5 tasks

One part first; then every share is one multiplication away.

- Share 720 in the ratio 2 : 3 : 7.
- If $A : B = 3 : 5$ and $B : C = 2 : 3$, find $A : B : C$.
- A mixture is 1 : 3 : 6 and uses 40 g of the first ingredient. Find the total mass.
- Three shares are in the ratio 4 : 5 : 6 and the largest is 42. Find the other two.
- Concrete 1 : 2 : 4 is needed to a total of 1 400 kg. Find each ingredient.